# Agent-Based Modeling of Rift Valley Fever Virus Transmission:

## A Scientific Literature Review with Focus on the Ferlo Ecosystem

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**Abstract**

Rift Valley fever (RVF) is a mosquito-borne zoonotic disease that poses significant threats to livestock and human health across Africa. This systematic literature review examines the application of agent-based modeling (ABM) and related computational approaches to understanding RVF virus (RVFV) transmission dynamics, with particular emphasis on studies integrating the Ferlo pastoral ecosystem of northern Senegal. Through analysis of 30 highly relevant studies, this review identifies key methodological approaches, transmission drivers, and ecosystem-specific insights. The Ferlo region emerges as a critical natural laboratory where temporary pond networks, nomadic herd movements, and seasonal rainfall patterns create unique transmission dynamics. Agent-based models developed for this ecosystem have revealed that herd mobility and precipitation are primary drivers of RVF outbreaks, with infection rates varying by up to 33% based on mobility patterns. The review synthesizes findings across modeling paradigms—from pure ABM to hybrid mechanistic-statistical approaches—and highlights the integration of remote sensing, hydrology, and vector ecology in contemporary RVF modeling. Key gaps include limited validation data, challenges in representing vertical transmission in mosquito vectors, and the need for more sophisticated behavioral models of pastoral livestock management.

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# Contents

**1 Introduction** . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . **4**

**2 Background and Theoretical Foundations** . . . . . . . . . . . . . . . . **5**

2.1 Rift Valley Fever Epidemiology . . . . . . . . . . . . . . . . . . . . . 5

2.2 The Rationale for Agent-Based Modeling . . . . . . . . . . . . . . . . . 5

**3 Modeling Approaches and Methodologies** . . . . . . . . . . . . . . . . **6**

3.1 Pure Agent-Based Models . . . . . . . . . . . . . . . . . . . . . . . . 6

3.2 Hybrid and Mechanistic Models . . . . . . . . . . . . . . . . . . . . . 6

3.3 Statistical and Bayesian Approaches . . . . . . . . . . . . . . . . . . . 7

3.4 Hierarchical Network Models . . . . . . . . . . . . . . . . . . . . . . . 7

3.5 Systematic Reviews of Modeling Approaches . . . . . . . . . . . . . . . 8

**4 The Ferlo Ecosystem: A Natural Laboratory for RVF Research** . **8**

4.1 Ecological and Climatic Characteristics . . . . . . . . . . . . . . . . . 8

4.2 Temporary Pond Networks and Vector Ecology . . . . . . . . . . . . . 8

4.3 Livestock Management and Herd Mobility . . . . . . . . . . . . . . . . 9

4.4 Endemic Circulation and Serological Evidence . . . . . . . . . . . . . 10

**5 Key Transmission Dynamics and Risk Factors** . . . . . . . . . . . . **10**

5.1 Rainfall and Hydrological Drivers . . . . . . . . . . . . . . . . . . . . 10

5.2 Herd Mobility and Contact Patterns . . . . . . . . . . . . . . . . . . . 11

5.3 Vector Population Dynamics and Species Complementarity . . . . . . 11
5.4 Vertical Transmission and Virus Persistence . . . . . . . . . . . . . . 12
5.5 Host Immunity and Population Dynamics . . . . . . . . . . . . . . . 12

**6 Integration of Environmental and Remote Sensing Data . . . . . . 13**
6.1 Satellite-Based Monitoring of Vector Habitats . . . . . . . . . . . . . 13
6.2 Hydrological Modeling . . . . . . . . . . . . . . . . . . . . . . . . . . 13
6.3 Climate Data and Forecasting . . . . . . . . . . . . . . . . . . . . . . 14

**7 Comparative Analysis of Model Findings . . . . . . . . . . . . . . . 14**
7.1 Convergent Evidence on Key Drivers . . . . . . . . . . . . . . . . . . 14
7.2 Divergent Findings and Unresolved Questions . . . . . . . . . . . . . 15
7.3 Model Validation and Uncertainty . . . . . . . . . . . . . . . . . . . . 16

**8 Discussion and Synthesis . . . . . . . . . . . . . . . . . . . . . . . . . 16**
8.1 The Value of Agent-Based Approaches . . . . . . . . . . . . . . . . . 16
8.2 The Ferlo Ecosystem as a Model System . . . . . . . . . . . . . . . . 17
8.3 Integration of Multiple Data Streams . . . . . . . . . . . . . . . . . . 17
8.4 From Understanding to Action . . . . . . . . . . . . . . . . . . . . . . 18

**9 Future Directions and Research Gaps . . . . . . . . . . . . . . . . . 18**
9.1 Methodological Advances . . . . . . . . . . . . . . . . . . . . . . . . . 18
9.2 Data Needs . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 19
9.3 Operational Applications . . . . . . . . . . . . . . . . . . . . . . . . . 20
9.4 Expanding Geographic Scope . . . . . . . . . . . . . . . . . . . . . . . 20
9.5 Climate Change Implications . . . . . . . . . . . . . . . . . . . . . . . 20

**10 Conclusion . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 21**

# 1. Introduction

Rift Valley fever (RVF) is a vector-borne viral disease that represents a major public health and economic challenge across sub-Saharan Africa and the Arabian Peninsula. Caused by a phlebovirus in the family *Phenuiviridae*, RVF primarily affects domestic ruminants but can cause severe disease in humans, ranging from mild febrile illness to hemorrhagic fever, encephalitis, and death. The disease's epidemic nature, driven by complex interactions between climatic conditions, vector ecology, and livestock management practices, has motivated extensive modeling efforts to understand transmission dynamics and inform control strategies.

Agent-based modeling (ABM) has emerged as a particularly valuable tool for studying RVF transmission because it can explicitly represent the spatial heterogeneity, individual-level behaviors, and stochastic processes that characterize disease spread in pastoral systems. Unlike traditional compartmental models that assume homogeneous mixing, ABM allows researchers to simulate the movements of individual animals, the spatial distribution of vector breeding sites, and the resulting contact patterns that drive transmission.

The Ferlo region of northern Senegal has become a focal point for RVF research due to its endemic circulation of the virus and the availability of detailed ecological and epidemiological data. This semi-arid Sahelian ecosystem, characterized by temporary pond networks and transhumant livestock systems, provides an ideal setting for studying the interplay between environmental drivers, vector dynamics, and host mobility in RVF transmission.

This review synthesizes the current state of knowledge on agent-based and related modeling approaches for RVF transmission, with particular attention to studies that have integrated data from the Ferlo ecosystem. We examine methodological innovations, key findings on transmission dynamics, the role of environmental factors, and identify critical gaps that should guide future research.

# 2. Background and Theoretical Foundations

## 2.1. Rift Valley Fever Epidemiology

RVF is maintained in nature through a complex transmission cycle involving multiple mosquito species and mammalian hosts. The primary vectors belong to the genera *Aedes* and *Culex*, with *Aedes vexans* and *Culex poicilipes* identified as key vectors in West African settings [Biteye et al., 2019]. *Aedes* mosquitoes are particularly important because they can transmit the virus vertically (transovarially) to their offspring, potentially allowing the virus to persist through inter-epidemic periods in desiccation-resistant eggs.

The disease exhibits a characteristic epidemic pattern linked to heavy rainfall events that create extensive breeding habitats for floodwater mosquitoes. During inter-epidemic periods, the virus circulates at low levels, maintained either through vertical transmission in mosquito populations or through continuous low-level transmission among livestock. Understanding the relative contributions of these maintenance mechanisms remains a central question in RVF epidemiology [Cécilia et al., 2022].

## 2.2. The Rationale for Agent-Based Modeling

Traditional compartmental models (SIR, SEIR, and their variants) have provided valuable insights into RVF dynamics but face limitations when representing the spatial and behavioral heterogeneity characteristic of pastoral systems. Agent-based models address these limitations by representing individual entities (animals, mosquitoes, humans) as autonomous agents with specific attributes and behaviors. This approach enables explicit modeling of:

- Spatial heterogeneity in vector breeding sites and host distributions
- Individual animal movements and herd management practices
- Contact networks arising from shared use of water points
- Stochastic transmission events at the individual level

- Heterogeneous immunity profiles within populations

The computational demands of ABM have historically limited their application, but advances in computing power and modeling platforms have made them increasingly feasible for vector-borne disease research [Paul et al., 2014].

# 3. Modeling Approaches and Methodologies

## 3.1. Pure Agent-Based Models

The foundational work on ABM for RVF in the Ferlo ecosystem was conducted by Paul and colleagues, who developed a multi-agent system specifically designed to study the impact of herd mobility on disease spread. Their 2014 model utilized the CORMAS (Common-pool Resources and Multi-Agent Systems) platform to simulate host-vector infection processes, animal herd mobility, and environmental factors including rainfall [Paul et al., 2014]. The model employed a variable-dimension grid representing approximately 100 square metres per cell, with daily time steps and typical simulation periods of 30 days. Key components included explicit representation of villages, ponds, pastures, and the movements of livestock herds between these locations.

This modeling framework was subsequently extended to examine the specific impact of rainfall on RVF transmission in the Ferlo region [Paul et al., 2018]. The agent-based approach proved particularly valuable for capturing the spatial dynamics of infection as herds moved between water points and the resulting contact patterns with mosquito populations concentrated around temporary ponds.

## 3.2. Hybrid and Mechanistic Models

While pure ABM approaches excel at representing spatial and behavioral heterogeneity, many recent studies have adopted hybrid frameworks that combine agent-based components with mechanistic or compartmental elements. Durand et al. [2020] developed a realistic epidemiological system model at the pond level that integrated vector population dynamics, resident and nomadic ruminant herd dynamics, and recorded herd movements in the Younoufere area of northern Senegal. This model employed a deterministic framework with discrete daily time steps, incorporating two closed vector populations (*Culex poicilipes* and *Aedes vexans*) and four host populations (cattle and small ruminants from both resident and nomadic herds).

The mechanistic modeling approach has also been applied to vector population dynamics specifically. Soti et al. [2012] combined hydrology and mosquito population models to identify drivers of RVF emergence in semi-arid West African regions. Their framework integrated a hydrological model predicting pond dynamics with an entomological model of mosquito population growth, providing a mechanistic link between rainfall, habitat availability, and vector abundance.

### 3.3. Statistical and Bayesian Approaches

Complementing mechanistic models, statistical approaches have been employed to analyze RVF incidence patterns and quantify uncertainty. Angelakis et al. [2025] applied Bayesian methods to model sparse RVF incidence data using zero-inflated self-exciting and autoregressive models. These approaches are particularly valuable for regions with limited surveillance data and can help identify spatial and temporal clustering of cases.

### 3.4. Hierarchical Network Models

Recognizing that RVF transmission occurs across multiple spatial scales, several studies have developed hierarchical network approaches. Xue et al. [2013] proposed a hierarchical network model for RVF epidemics with applications in North America, representing transmission within local populations (nodes) and between populations through animal movements (edges). This framework bridges the gap between local transmission dynamics and regional epidemic spread.

### 3.5. Systematic Reviews of Modeling Approaches

Cécilia et al. [2022] conducted a comprehensive systematic review of mechanistic models of RVF virus transmission, identifying 37 relevant studies published between 1980 and 2021. Their analysis revealed that most models focused on vector-host transmission dynamics, with fewer incorporating environmental drivers or spatial heterogeneity. The review highlighted the diversity of modeling approaches but also noted limited model validation and comparison, underscoring the need for standardized frameworks and data-sharing practices.

## 4. The Ferlo Ecosystem: A Natural Laboratory for RVF Research

### 4.1. Ecological and Climatic Characteristics

The Ferlo region of northern Senegal represents a semi-arid Sahelian ecosystem characterized by distinct wet and dry seasons. Annual rainfall ranges from 300 to 500 mm, concentrated in a rainy season extending from July to October or November [Paul et al., 2014, Durand et al., 2020]. This seasonal precipitation pattern drives the formation of a complex network of temporary ponds (*mares*) that fill during the rains and progressively dry out during the dry season.

The vegetation consists of various types adapted to the semi-arid conditions, providing grazing resources for livestock. The region supports a mixed livestock system including cattle, buffalo, sheep, goats, and camels, managed through both sedentary and transhumant pastoral practices [Paul et al., 2014].

### 4.2. Temporary Pond Networks and Vector Ecology

The temporary ponds of the Ferlo serve multiple critical functions in RVF transmission ecology. They provide breeding habitats for mosquito vectors, particularly *Aedes vexans* and *Culex poicilipes*, which are the primary vectors in this region [Durand et al., 2020]. The ponds also serve as congregation points for livestock, creating spatial foci where vector-host contact rates are elevated.

Tourre et al. [2008] mapped zones potentially occupied by *Aedes vexans* and *Culex poicilipes* mosquitoes in Senegal, identifying the spatial distribution of vector habitats and their relationship to environmental variables. This spatial mapping provides essential input data for spatially explicit transmission models.

Biteye et al. [2019] investigated the host-feeding patterns of *Aedes vexans arabiensis* in the Ferlo pastoral ecosystem, finding that this species feeds opportunistically on available hosts, with implications for virus amplification and spillover to humans. Understanding vector feeding behaviour is crucial for parameterising transmission models and assessing zoonotic risk.

### 4.3. Livestock Management and Herd Mobility

A defining feature of the Ferlo ecosystem is the coexistence of resident and nomadic livestock herds. Resident herds remain within relatively circumscribed areas year-round, while nomadic herds undertake seasonal movements (transhumance) in response to forage and water availability. These movement patterns have profound implications for RVF transmission dynamics.

Paul et al. [2014] demonstrated that herd mobility significantly influences infection rates, with a 33% variation observed between scenarios with no mobility and high mobility. Increased mobility, particularly movement toward water sources during the dry season, concentrates animals around ponds where vector densities are highest, thereby amplifying transmission.

The timing and routes of nomadic movements also affect disease dynamics. Durand et al. [2020] found that nomadic herds arriving during the second half of the rainy season had higher predicted proportions of viraemic individuals (3.4%), consistent with amplification of virus circulation during this period.

### 4.4. Endemic Circulation and Serological Evidence

Unlike the explosive epidemics observed in East Africa, RVF in the Ferlo region exhibits endemic circulation with periodic amplification during favourable conditions. Serological surveys conducted by Durand et al. [2020] in 2015–2016 confirmed long-lasting RVF endemicity in resident herds (IgG seroprevalence of 15.3%, $n = 222$) and provided the first estimates of seroprevalence in nomadic herds in West Africa (12.4%, $n = 660$). The detection of IgM antibodies in 4% of nomadic herds sampled during the second half of the rainy season indicated recent or ongoing transmission.

Multivariate analysis of serological data suggested amplification of the transmission cycle during the rainy season with a peak of circulation at the end of that season [Durand et al., 2020]. Age was identified as a significant risk factor for seropositivity in small ruminants, reflecting cumulative exposure over time.

## 5. Key Transmission Dynamics and Risk Factors

### 5.1. Rainfall and Hydrological Drivers

Rainfall emerges consistently across studies as a primary driver of RVF transmission. The relationship operates through multiple pathways: rainfall creates and maintains mosquito breeding habitats, influences vegetation growth and livestock distribution, and affects the timing and extent of herd movements.

Paul et al. [2014] found that strong rains amplify transmission by increasing pond volumes, leading to mosquito multiplication and higher vector populations. Controlling rainfall amounts to manage water ponds was identified as crucial, as dry ponds reduce herd contact and infection risk. Their simulations demonstrated that the evolution of infected hosts and vectors depends critically on precipitation patterns.

Soti et al. [2012] developed an integrated hydrology-mosquito model that mechanistically linked rainfall to pond dynamics and vector abundance. Their model identified specific hydrological conditions that favour RVF emergence in semi-arid regions, providing a quantitative framework for early warning systems.

Chemison et al. [2024] demonstrated that a dynamical climate-sensitive disease model could reproduce historical RVF outbreaks across Africa, with epidemic severity strongly correlating with rainy season duration. Epidemics were observed to terminate abruptly when rain stopped, highlighting the tight coupling between environmental conditions and transmission.

### 5.2. Herd Mobility and Contact Patterns

Animal movement is a second critical driver of RVF transmission, operating at multiple scales. At the local scale, daily movements of herds to water points create predictable spatial foci of transmission. At the regional scale, nomadic movements can introduce virus into new areas or maintain endemic circulation through repeated introductions.

The agent-based model of Paul et al. [2014] explicitly demonstrated that increased herd mobility, especially movement toward water sources, significantly raises infection rates. This finding has important implications for control strategies, suggesting that interventions targeting high-contact locations (water points) or modifying movement patterns could reduce transmission.

Durand et al. [2020] used model comparison techniques to evaluate different scenarios of virus introduction and maintenance. Their results demonstrated that nomadic movements alone are sufficient to account for endemic circulation in northern Senegal, with yearly introductions of RVFV by nomadic herds explaining observed seroprevalence patterns.

### 5.3. Vector Population Dynamics and Species Complementarity

The Ferlo ecosystem supports multiple mosquito vector species with distinct ecological niches and transmission roles. *Aedes vexans* is a floodwater mosquito that emerges in large numbers following rainfall, while *Culex poicilipes* breeds in more permanent water bodies and maintains populations throughout the year.

This species complementarity has important implications for transmission dynamics. *Aedes* species drive explosive amplification during the rainy season, while *Culex* species may maintain low-level transmission during inter-epidemic periods. Durand et al. [2020] incorporated both vector populations in their model, representing them as closed populations with distinct dynamics.

Drouin et al. [2024] developed mechanistic models of mosquito population dynamics for multiple potential vector species in the western Mediterranean Basin, including *Aedes vexans*. Their daily time-step model with 0.1° spatial resolution provided spatially explicit forecasts of vector abundance, demonstrating the feasibility of operational vector surveillance systems.

### 5.4. Vertical Transmission and Virus Persistence

The mechanism by which RVF virus persists between epidemics remains incompletely understood. Vertical transmission in *Aedes* mosquitoes, where infected females pass the virus to their eggs, has been proposed as a key persistence mechanism. However, empirical evidence for the efficiency and epidemiological significance of this pathway is limited.

Durand et al. [2020] explicitly tested scenarios with and without vertical transmission in *Aedes vexans*. While their model could not definitively rule out vertical transmission, the results demonstrated that nomadic herd movements alone were sufficient to explain endemic circulation, suggesting that continuous reintroduction may be more important than local persistence in the Ferlo context.

### 5.5. Host Immunity and Population Dynamics

Naturally acquired immunity following infection influences both individual susceptibility and population-level transmission dynamics. Chemison et al. [2024] found that herd immunity affects total mortality over multi-year periods. The accumulation of immune individuals in a population reduces the effective reproductive number of the virus, potentially preventing or limiting epidemic spread.

Age-structured immunity patterns observed in serological surveys reflect the history of virus circulation. Durand et al. [2020] identified age as a significant risk factor for seropositivity in small ruminants, consistent with cumulative exposure over time.

## 6. Integration of Environmental and Remote Sensing Data

### 6.1. Satellite-Based Monitoring of Vector Habitats

Remote sensing technologies have become integral to RVF modeling and surveillance, providing spatially explicit data on environmental conditions that drive vector population dynamics. Vignolles et al. [2010] developed an operational warning system for RVF risk using TerraSAR-X high-resolution radar remote sensing. This system detects temporary ponds and monitors their dynamics, providing critical input for models of mosquito abundance and transmission risk.

The integration of multiple satellite platforms and sensors has enhanced the spatial and temporal resolution of environmental monitoring. Tran et al. [2019] integrated satellite-derived meteorological estimates in population dynamics models for mosquito vectors in northern Senegal. This approach allows near-real-time updating of model predictions based on observed environmental conditions.

### 6.2. Hydrological Modeling

Accurate representation of pond dynamics is essential for modeling RVF transmission in semi-arid regions where temporary water bodies drive both vector ecology and livestock distribution. Soti et al. [2012] developed a coupled hydrology-mosquito model that mechanistically linked rainfall to pond formation, persistence, and mosquito population growth. Their model identified specific hydrological conditions associated with RVF emergence, providing quantitative thresholds for early warning.

Lafaye et al. [2013] described the AdaptFVR project, which aimed to develop proactive adaptation strategies for livestock management in Senegal through integration of hydrological, entomological, and epidemiological data. This project exemplifies the multi-disciplinary approach required for operational RVF risk management.

### 6.3. Climate Data and Forecasting

Climate variability, particularly rainfall patterns, is a primary driver of RVF epidemic risk. Integration of climate data from sources such as the Tropical Rainfall Measuring Mission (TRMM) has improved model accuracy and enabled forecasting of high-risk periods.

McMahon et al. [2014] coupled vector-host dynamics with weather, geography, and mitigation measures to model RVF in Africa. Their model incorporated spatially explicit climate data and demonstrated how geographic variation in environmental conditions influences transmission potential across the continent.

Chemison et al. [2024] developed a dynamical climate-sensitive disease model that successfully reproduced historical RVF outbreaks across Africa over a 25-year period. The model's ability to capture observed epidemic patterns validates the central role of climate drivers and suggests potential for operational forecasting systems.

## 7. Comparative Analysis of Model Findings

### 7.1. Convergent Evidence on Key Drivers

Despite diversity in modeling approaches and geographic focus, several findings emerge consistently across studies:

1. **Rainfall is the primary environmental driver**: All models incorporating environmental variables identify rainfall as the dominant driver of RVF transmission, operating through effects on vector breeding habitats, vegetation, and livestock distribution [Paul et al., 2014, 2018, Soti et al., 2012, Chemison et al., 2024, McMahon et al., 2014].

2. **Herd mobility amplifies transmission**: Studies explicitly modeling animal movements consistently find that mobility increases infection risk, particularly when movements concentrate animals at water points [Paul et al., 2014, Durand et al., 2020].
3. **Seasonal amplification patterns**: Multiple studies identify amplification of transmission during the rainy season with peak circulation toward the end of the season [Durand et al., 2020, Chemison et al., 2024].
4. **Vector abundance as a proximate driver**: Models incorporating vector dynamics confirm that mosquito abundance is a key proximate determinant of transmission intensity [Drouin et al., 2024, Soti et al., 2012].

### 7.2. Divergent Findings and Unresolved Questions

Several important questions remain unresolved or show divergent findings across studies:

1. **Relative importance of vertical transmission**: While vertical transmission in *Aedes* mosquitoes is biologically plausible and has been demonstrated experimentally, its epidemiological significance remains uncertain. Durand et al. [2020] found that nomadic movements alone could explain endemic circulation without invoking vertical transmission, while other studies have assumed vertical transmission is necessary for inter-epidemic persistence.
2. **Optimal control strategies**: Different models have evaluated various control interventions, including vector control, movement restrictions, and vaccination, but consensus on optimal strategies for different epidemiological contexts is lacking [Chemison et al., 2024, McMahon et al., 2014].
3. **Spatial scale of transmission**: The appropriate spatial scale for modeling RVF transmission—from individual ponds to regional landscapes—remains debated, with implications for data requirements and computational feasibility.

### 7.3. Model Validation and Uncertainty

A critical limitation identified across the literature is limited model validation against independent data. Cécilia et al. [2022] noted that few models have been rigorously validated against observed epidemic patterns. This reflects both the scarcity of detailed epidemiological data and the challenges of validating stochastic, spatially explicit models.

Durand et al. [2020] employed Adaptive Population Monte-Carlo Approximate Bayesian Computation (ABC-APMC) methods for parameter estimation, providing a rigorous framework for quantifying uncertainty. However, such approaches remain the exception rather than the norm in RVF modeling.

## 8. Discussion and Synthesis

### 8.1. The Value of Agent-Based Approaches

The studies reviewed demonstrate the particular value of agent-based modeling for RVF transmission in pastoral systems. By explicitly representing individual animals, herds, and their movements, ABM captures spatial and behavioral heterogeneity that fundamentally shapes transmission dynamics. The work of Paul et al. in the Ferlo ecosystem exemplifies how ABM can reveal non-intuitive relationships, such as the quantitative impact of herd mobility on infection rates [Paul et al., 2014, 2018].

However, ABM also presents challenges. These models are data-intensive, requiring detailed information on animal movements, vector distributions, and environmental conditions. They are computationally demanding, limiting the spatial and temporal scales that can be simulated. And they involve numerous parameters that must be estimated or calibrated, introducing uncertainty.

### 8.2. The Ferlo Ecosystem as a Model System

The Ferlo region has emerged as a critical natural laboratory for RVF research, offering several advantages:

1. **Endemic circulation**: Unlike regions experiencing sporadic epidemics, the Ferlo exhibits continuous low-level circulation, enabling study of maintenance mechanisms [Durand et al., 2020].
2. **Well-characterised ecology**: Extensive research has documented the region's hydrology, vector ecology, and livestock systems, providing rich data for model parameterisation [Tourre et al., 2008, Vignolles et al., 2010, Soti et al., 2012, Lafaye et al., 2013, Biteye et al., 2019].
3. **Coexistence of resident and nomadic systems**: The interaction between resident and nomadic herds provides a natural experiment for studying the role of animal movements in transmission [Durand et al., 2020].
4. **Availability of serological data**: Repeated serological surveys have quantified seroprevalence in both resident and nomadic populations, enabling model validation [Durand et al., 2020].

Insights from the Ferlo ecosystem have broader applicability to other semi-arid pastoral regions of the Sahel and East Africa where similar ecological and management conditions prevail.

### 8.3. Integration of Multiple Data Streams

A key trend in recent RVF modeling is the integration of multiple data streams—remote sensing, climate data, entomological surveys, serological data, and animal movement data—into unified modeling frameworks. This integration is exemplified by the work of Soti et al. [2012], who coupled hydrological and entomological models, and the AdaptFVR project described by Lafaye et al. [2013], which aimed to integrate diverse data sources for operational risk management.

The increasing availability of high-resolution satellite data, automated weather stations, and GPS tracking of livestock movements promises to further enhance model realism and predictive capability. However, realising this potential requires sustained investment in data collection infrastructure and capacity building in affected regions.

### 8.4. From Understanding to Action

Ultimately, the value of RVF modeling lies in its ability to inform prevention and control strategies. Several studies have explicitly evaluated intervention scenarios:

- Chemison et al. [2024] found that vector control, movement control, and annual vaccination of young animals can significantly reduce cattle mortality, even without expected rainfall.
- McMahon et al. [2014] explored how mitigation measures interact with geographic and climatic variation to influence control effectiveness.
- Paul et al. [2014] suggested that controlling rainfall amounts to manage water ponds could reduce transmission, though the practical feasibility of such interventions is limited.

Translating model insights into operational public health and veterinary interventions requires close collaboration between modelers, public health authorities, and livestock keepers. Participatory modeling approaches that engage stakeholders in model development and scenario evaluation may enhance the uptake and impact of modeling results.

## 9. Future Directions and Research Gaps

### 9.1. Methodological Advances

Several methodological advances would enhance RVF modeling:

1. **Improved representation of vector ecology**: Most models treat vector populations in simplified ways. More mechanistic representation of mosquito

life cycles, species-specific behaviors, and vertical transmission would improve realism [Drouin et al., 2024, Soti et al., 2012].

2. **Integration of human dimensions**: While livestock are central to RVF transmission, human cases represent the ultimate public health concern. Few models explicitly represent human populations and zoonotic spillover [Paul et al., 2014].
3. **Behavioral models of livestock management**: Pastoral livestock management involves complex decision-making about movements, herd composition, and veterinary interventions. Incorporating these behavioral dimensions could improve model realism and enable evaluation of behavior-change interventions [Iles et al., 2020].
4. **Multi-scale modeling frameworks**: Transmission occurs across scales from individual ponds to regional landscapes. Hierarchical or multi-scale modeling approaches that link these scales are needed [Xue et al., 2013].

### 9.2. Data Needs

Critical data gaps limit model development and validation:

1. **Longitudinal vector abundance data**: Time series of mosquito abundance at multiple sites would enable better parameterisation and validation of vector population models [Drouin et al., 2024, Soti et al., 2012].
2. **Animal movement data**: GPS tracking of livestock herds would provide empirical data on movement patterns to replace assumptions in current models [Paul et al., 2014, Durand et al., 2020].
3. **Viral circulation data**: Virological surveillance to detect circulating virus in vectors and hosts during inter-epidemic periods would clarify persistence mechanisms [Durand et al., 2020].
4. **Experimental transmission data**: Laboratory and field studies quantifying transmission efficiency between vectors and hosts would reduce parameter uncertainty [Bron et al., 2023].

### 9.3. Operational Applications

Translating research models into operational tools requires:

1. **Real-time data integration**: Systems that automatically ingest satellite, climate, and surveillance data to update risk predictions [Vignolles et al., 2010, Lafaye et al., 2013].
2. **User-friendly interfaces**: Decision support tools that present model outputs in accessible formats for non-specialist users.
3. **Validation and performance monitoring**: Systematic evaluation of model predictions against observed outcomes to build confidence and identify areas for improvement.
4. **Capacity building**: Training of local researchers and public health professionals in modeling methods and interpretation.

### 9.4. Expanding Geographic Scope

While the Ferlo ecosystem has been intensively studied, other regions affected by RVF would benefit from similar modeling efforts. The Mediterranean Basin, where competent vectors are present and climate change may increase transmission risk, represents a priority area [Drouin et al., 2024]. East African countries experiencing periodic explosive epidemics could benefit from models tailored to their specific ecological and management contexts [Chemison et al., 2024].

### 9.5. Climate Change Implications

Climate change is expected to alter rainfall patterns, temperature regimes, and land use in RVF-affected regions. Models that incorporate climate projections could help anticipate how transmission risk may shift geographically and temporally. Chemison et al. [2024] provide a foundation for such projections, but more work is needed to quantify uncertainty and identify adaptation strategies.

## 10. Conclusion

Agent-based modeling has proven to be a valuable tool for understanding Rift Valley fever virus transmission dynamics, particularly in complex pastoral systems where spatial heterogeneity and individual behaviors shape disease spread. The body of work reviewed here demonstrates that rainfall and herd mobility are primary drivers of RVF transmission, operating through their effects on vector breeding habitats, mosquito abundance, and contact patterns between vectors and hosts.

The Ferlo ecosystem of northern Senegal has emerged as a critical natural laboratory where endemic RVF circulation, well-characterised ecology, and the coexistence of resident and nomadic livestock systems enable detailed study of transmission mechanisms. Research in this region has revealed that nomadic herd movements are sufficient to maintain endemic circulation without necessarily invoking vertical transmission in mosquito vectors, though the relative importance of these mechanisms may vary across settings.

Integration of remote sensing, hydrological modeling, and entomological data has enhanced model realism and enabled development of early warning systems. However, significant gaps remain, including limited validation data, uncertainty about vertical transmission efficiency, and incomplete understanding of human behavioral factors in livestock management.

Future advances will require sustained investment in data collection infrastructure, methodological innovations in multi-scale and behavioral modeling, and close collaboration between modelers and public health practitioners to translate insights into effective interventions. As climate change alters the environmental conditions that drive RVF transmission, modeling will play an increasingly important role in anticipating risk and guiding adaptation strategies.

The convergence of evidence across diverse modeling approaches provides confidence in key findings while highlighting areas where further research is needed. By continuing to refine models, collect validation data, and engage stakeholders in participatory modeling processes, the research community can enhance the contribution of modeling to RVF prevention and control.

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